

CSA5112 - Cryptography and Network Security

9. Double DES (2DES)

Aim: To implement the Double DES (2DES) algorithm for encrypting and decrypting a plaintext message using two different DES keys.

Algorithm:

1. Start.
2. Import the required DES modules.
3. Read the plaintext.
4. Define two different 8-byte DES keys.
5. Create two DES cipher objects.
6. Encrypt the plaintext using the first DES key.
7. Encrypt the resulting ciphertext using the second DES key.
8. Display the final ciphertext.
9. Decrypt the ciphertext using the second key.
10. Decrypt the result using the first key.
11. Display the original plaintext.
12. Stop.

Code:

```
from Crypto.Cipher import DES

from Crypto.Util.Padding import pad, unpad

import base64

# Two DES Keys (8 bytes each)

key1 = b'12345678'

key2 = b'ABCDEFGH'

cipher1 = DES.new(key1, DES.MODE_ECB)

cipher2 = DES.new(key2, DES.MODE_ECB)
```

```
plaintext = input("Enter Plain Text: ")

# First Encryption

step1 = cipher1.encrypt(pad(plaintext.encode(), DES.block_size))

# Second Encryption

step2 = cipher2.encrypt(step1)

ciphertext = base64.b64encode(step2).decode()

print("\nEncrypted Text:", ciphertext)

# First Decryption (reverse order)

decrypt1 = cipher2.decrypt(step2)

# Second Decryption

decrypt2 = unpad(cipher1.decrypt(decrypt1), DES.block_size)

print("Decrypted Text:", decrypt2.decode())
```

Sample Input:

```
Enter Plain Text: HELLO
```

Sample Output:

```
Enter Plain Text: HELLO

Encrypted Text: J+3Qb2I1d7GvXvRZ7o2+6Q==
Decrypted Text: HELLO
```

Result:

The Double DES (2DES) algorithm was successfully implemented. The plaintext was encrypted using two different DES keys and successfully decrypted back to the original plaintext.